INTRODUCTION

We will now use discounted cash flow analysis to calculate bond prices and show how the discount rate used as well as a bond's features, such as its coupon rate and time-to-maturity, affect pricing. The price of a bond and its future cash flows can be used calculate an internal rate of return, known as the yield-to-maturity, which serves as a useful return measure for fixed-income investors under certain assumptions. One of these assumptions that applies to this learning module is that all bond interest and principal cash flows occur as promised. We will explore the relationship between bond prices and bond features, showing how different features affect a bond's price, and demonstrate pricing both on and between bond coupon dates. Finally, we will introduce matrix pricing, which uses comparable bonds to estimate a bond's price and yield-to-maturity when neither is known.

Most of the examples and exhibits used throughout the reading can be downloaded as a Microsoft Excel workbook. Each worksheet in the workbook is labeled with the corresponding example or exhibit number in the text.

Learning Module Overview

- Bond pricing is an application of discounted cash flow analysis. Bond prices are a function of a bond's features, including its cash flows and the interest rate(s) used to discount future cash flows.
- By comparing a bond's price to its face value or its coupon rate to the discount rate, we can identify whether a bond is trading at a discount, at par, or at a premium.
- If the market price of a bond is known, an internal rate of return on the cash flows can be calculated, known as the yield-to-maturity (YTM). The YTM is the single interest rate that equates the present value of future cash flows to the price of the bond.
- A bond investor's rate of return will equal the YTM if (1) the investor holds the bond to maturity, (2) the issuer makes full coupon and principal payments on the scheduled dates, and (3) the investor reinvests all coupon payments at the same YTM.
- When a bond is priced or traded in between coupon dates, an additional amount must be added for interest that has accrued since the last coupon payment, to compensate the seller, since the buyer will receive the entire next coupon payment.
- To calculate a bond's accrued interest on any date, we multiply the coupon by the fraction of days elapsed in the coupon period divided by the total days in the coupon period. There are various conventions for counting these days; two common conventions are actual/actual and 30/360.
- A bond's price changes inversely with changes in its YTM. A bond's features determine price sensitivity to changes in YTM.
- The lower the coupon rate on a fixed-coupon bond, the greater the percentage price change for a given change in the bond's yield-to-maturity. Generally, the longer a bond's time-to-maturity, the greater its percentage price change for a given change in its yield-to-maturity.
- Unlike listed equity securities, most bonds are thinly traded, which complicates price discovery.

quotations for bonds.

Learning Module Self-Assessment

These initial questions are intended to help you gauge your current level of understanding of the learning module.

1. A corporate bond that matures on 1 January 2035 pays semiannual coupons of 3.25% per year and has a face value of 100. The market discount rate is 4.0%. For a trade settlement date of 1 January 2030, the price of the bond as a percentage of par value, assuming a 30/360 day count, is *closest* to:

A. 96.632.
B. 96.661.
C. 103.436.

Hide Solution

Solution:

A is correct. The bond price is the sum of the coupon and principal payments discounted at the market discount rate.

$$PV = PMT_1/(1 + r)^1 + PMT_2/(1 + r)^2 + \ldots + (PMT_N + FV_N)/(1 + r)^N,$$

where

$PMT = 1.625$

$r = 0.02$

$FV = 100$

$N = 10$

$$PV = \frac{1.625}{(1+0.02)^1} + \frac{1.625}{(1+0.02)^2} + \frac{1.625}{(1+0.02)^3} + \frac{1.625}{(1+0.02)^4} + \frac{1.625}{(1+0.02)^5} +$$

$$\frac{1.625}{(1+0.02)^6} + \frac{1.625}{(1+0.02)^7} + \frac{1.625}{(1+0.02)^8} + \frac{1.625}{(1+0.02)^9} + \frac{101.625}{(1+0.02)^{10}}$$

$$PV = 96.632.$$

Note: Quantities in the calculation are not rounded, though they are shown with three decimal places for presentation purposes. Solutions are rounded to three decimal places.

The calculation can also be done using the PV or PRICE functions in Microsoft Excel and Google Sheets:

```
=–PV(RATE, NPER, PMT, [FV], TYPE)
```

=-PV(0.02,10,1.625,100,0)

= 96.632

```
=PRICE(SETTLEMENT, MATURITY, RATE, YIELD, REDEMPTION, FREQUENCY,
[BASIS])
```

=PRICE(DATE(2030,1,1),DATE(2035,1,1),0.0325,0.04,100,2)

= 96.632

2. Which of the following is *not* required for a bond investor's rate of return on a bond investment to equal the bond's YTM?

C. Coupon payments are reinvested at the same rate as the YTM.

Hide Solution

Solution:

B is correct. For a bond investor to earn the YTM, the issuer must make all coupon and principal payments on their scheduled dates.

A is incorrect because a bond investor does have to hold the bond to maturity to earn the YTM.

C is incorrect because a bond investor does have to reinvest coupons at the same rate as the YTM to earn the YTM.

3. A three-year sovereign bond issued on 1 January 2030 pays semiannual coupons of 1.5% per year on 30 June and 31 December each year and has a face value of 100. The market discount rate is 2.0%. For a trade settlement date of 29 August 2031, the flat price of the bond as a percentage of par value, assuming an actual/actual day count, is *closest* to:

A. 99.343.
B. 99.587.
C. 99.832.

Hide Solution

Solution:

A is correct. The flat price is the full price less accrued interest. The full price is the present value as of the trade settlement date, which involves partial payment periods because we're between coupon dates.

$$PV^{Full} = \frac{PMT}{(1+r)^{1-t/T}} + \frac{PMT}{(1+r)^{2-t/T}} + \cdots + \frac{PMT+FV}{(1+r)^{N-t/T}}.$$

Here,

PMT = 0.75

r = 0.01

t = 60 (days from 30 June 2031 to 29 August 2031)

T = 184 (days from 30 June 2031 to next coupon on 31 December 2031)

FV = 100

N = 3

$$PV_{Full} = \frac{0.75}{(1+0.01)^{1-60/184}} + \frac{0.75}{(1+0.01)^{2-60/184}} \frac{100.75}{(1+0.01)^{3-60/184}}$$

$$PV_{Full} = 99.587.$$

Accrued interest is the proportional share of the next coupon payment owed the seller.

$$AI = \frac{t}{T} \times PMT,$$

$$AI = \frac{60}{184} \times 0.75$$

$$AI = 0.245.$$

The flat price is 99.587 – 0.245 = 99.343.

The present value calculations can also be done easily on spreadsheet software. Additionally, the PRICE function in Microsoft Excel and Google Sheets can be used to directly solve for the flat price (note: *not* full prices).

=PRICE(*SETTLEMENT, MATURITY, RATE, YIELD, REDEMPTION, FREQUENCY, [BASIS]*)

4. True or false: A bond's quoted price is also called its full price because it is the amount that the buyer pays the seller.

A. True
B. False

Hide Solution

Solution:

False. A bond is quoted by its flat price. The full price is the amount paid by the buyer to the seller, and it includes accrued interest if settlement does not occur on a coupon payment date.

5. Assume that the Japanese government issues two non-callable fixed-coupon bonds on the same date with the same coupon rate. The bonds are identical except that one matures in 10 years and the other matures in 30 years. If the relevant market discount rates for both bonds rise by 65 bps, which of the following will be true?

A. The 10-year bond will have a larger percentage price change than the 30-year bond.
B. The 10-year bond will have a smaller percentage price change than the 30-year bond.
C. The 10-year bond and the 30-year bond will have equal percentage price changes.

Hide Solution

Solution:

B is correct. Generally, for the same coupon rate, a shorter-term bond will have a smaller percentage price change than a longer-term bond when their relevant market discount rates change by the same amount.

A is incorrect because generally a shorter-term bond will have a smaller percentage price change than a longer-term bond when their market discount rates change by the same amount.

C is incorrect because shorter-term bonds and longer-term bonds will not have the same percentage price changes when their market discount rates change by the same amount.

6. True or false. Justify your answer.

Matrix pricing is an estimation process in which quotes from derivative markets (e.g., interest rates swaps) are used to determine the price of illiquid or new bonds for which transactional data are scarce.

A. True
B. False

Hide Solution

Solution:

False. Matrix pricing is a process for estimating prices of illiquid or newly issued bonds using prices of comparable bonds, not from transactions in derivative markets. The comparable bonds have similar times-to-maturity, coupon rates, and credit quality as those of the bond under study.

What do you want to discuss... (required)

NN Can someone please explain how to do 3 Question

Created 5 months ago by Navinya Nagarajan 3 replies | Last Activity: 2 months ago

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MG Could someone explain better how to do q3 in the calc BA II plus?

Created 5 months ago by Mariana González Vázquez 2 replies | Last Activity: 4 months ago

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KR Question set #3, shouldn't N= 6? since it is a semi annual coupon?

Created 5 months ago by kisho Riel Hagiwara 1 reply | Last Activity: 5 months ago

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